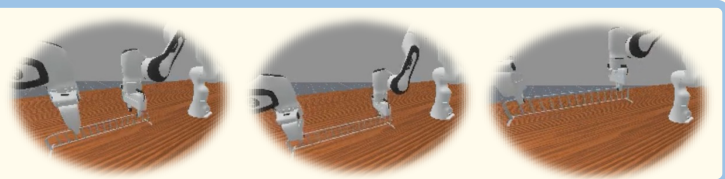
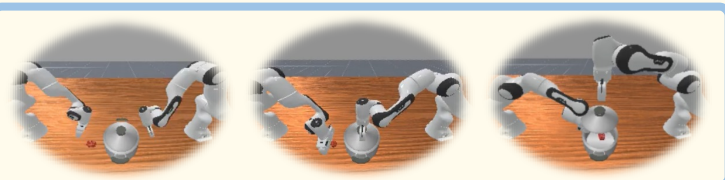


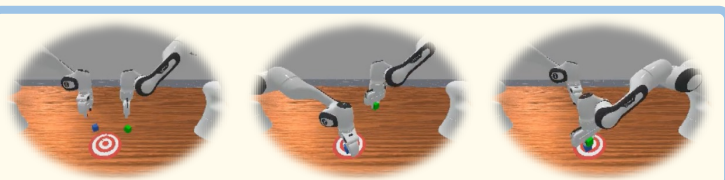
(a) Multi-Agent Manipulation Tasks



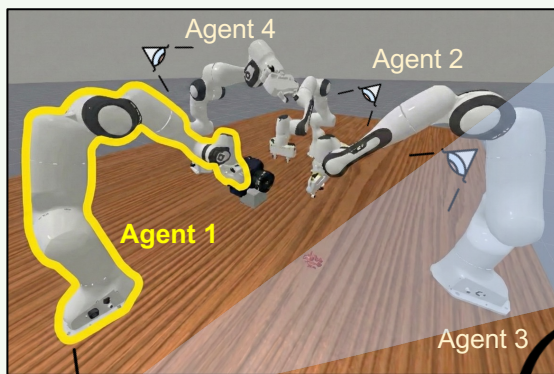
Tight synchronization



Role-asymmetric coordination



Strict sequential dependency



Shared global obs.



MA-dynamics



MA-Kin.
Encoder

Privileged
CLS

Distillation



MA-Kin.
Decoder

Contextualizer

ViT-B/16 SigLIP + Transformer



Vision
Encoder

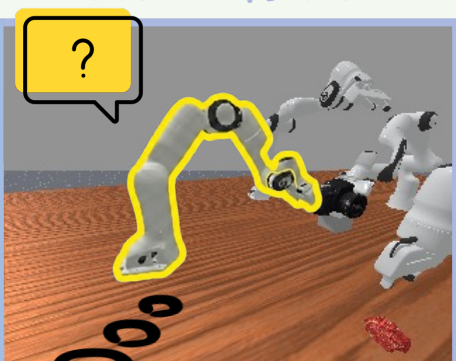
Local RGB obs.



Text
Encoder

Task Instruction

"Lift the object, align
camera with two arms,
and press the shutter."



Agent 1's local obs.

w/ partial observability

(b) CLS-DP's Model Architecture

Coordinated
Action Sequence

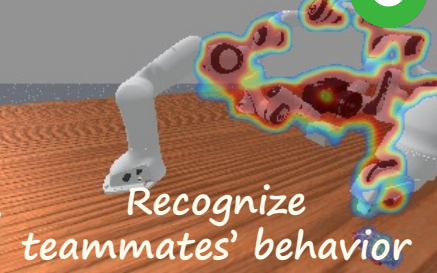


Action-Expert

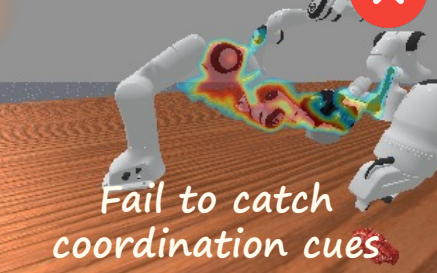
Decentralized Diffusion Policy



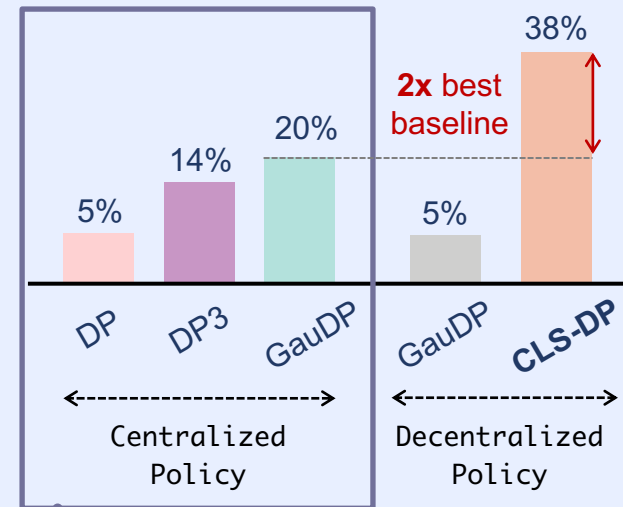
w/ CLS



w/o CLS



(c) Avg. Task Success Rates



For coordination,
centralized policy must employ

- All local obs.
- Shared global obs.
- All proprio. states

As the team
size increases,
causing

limited scalability
- Comm./sync. bottleneck
Failure vulnerability